

A. TECHNICAL TOOLS

A.1 Probabilistic Inequalities

A.1.1 Tail Bounds with Stopping Times

Lemma 33 (Hoeffding's inequality with adaptive stopping time): For i.i.d. random variables $Z_1, \dots, Z_T$ taking values in $[a, b]$ almost surely, with probability at least $1 - \delta$,

$$\frac{1}{T'} \sum_{i=1}^{T'} Z_i - \mathbb{E}[Z] \leq (b - a) \sqrt{\frac{\log(T/\delta)}{2T'}} \quad \forall 1 \leq T' \leq T. \quad (\text{A.1})$$

As a consequence, for any random variable $\tau \in [T]$ with the property that for all $t \in [T]$, $\mathbb{I}\{\tau \leq t\}$ is a measurable function of $Z_1, \dots, Z_{t-1}$ (τ is called a *stopping time*), we have that with probability at least $1 - \delta$,

$$\frac{1}{\tau} \sum_{i=1}^{\tau} Z_i - \mathbb{E}[Z] \leq (b - a) \sqrt{\frac{\log(T/\delta)}{2\tau}}. \quad (\text{A.2})$$

Proof of Lemma 33. Lemma 3 states that for any fixed $T' \in [T]$, with probability at least $1 - \delta$,

$$\frac{1}{T'} \sum_{i=1}^{T'} Z_i - \mathbb{E}[Z] \leq (b - a) \sqrt{\frac{\log(T/\delta)}{2T'}}.$$

(A.1) follows by applying this result with $\delta' = \delta/T$ and taking a union bound over all T choices for $T' \in [T]$. For (A.2), we observe that

$$\frac{1}{\tau} \sum_{i=1}^{\tau} (Z_i - \mathbb{E}[Z]) - (b - a) \sqrt{\frac{\log(T/\delta)}{2\tau}} \leq \max_{T' \in [T]} \left\{ \frac{1}{T'} \sum_{i=1}^{T'} (Z_i - \mathbb{E}[Z]) - (b - a) \sqrt{\frac{\log(T/\delta)}{2T'}} \right\}.$$

The result now follows from (A.1). $\square$

A.1.2 Tail Bounds for Martingales

Lemma 34: For any sequence of real-valued random variables $(X_t)_{t \leq T}$ adapted to a filtration $(\mathcal{F}_t)_{t \leq T}$, it holds that with probability at least $1 - \delta$, for all $T' \leq T$,

$$\sum_{t=1}^{T'} X_t \leq \sum_{t=1}^{T'} \log(\mathbb{E}_{t-1}[e^{X_t}]) + \log(\delta^{-1}). \quad (\text{A.3})$$

Proof of Lemma 34. We claim that the sequence

$$Z_\tau := \exp \left(\sum_{t=1}^{\tau} X_t - \log(\mathbb{E}_{t-1}[e^{X_t}]) \right)$$

is a nonnegative supermartingale with respect to the filtration $(\mathcal{F}_\tau)_{\tau \leq T}$. Indeed, for any choice of τ , we have

$$\begin{aligned}\mathbb{E}_{\tau-1}[Z_\tau] &= \mathbb{E}_{\tau-1} \left[\exp \left(\sum_{t=1}^{\tau} X_t - \log(\mathbb{E}_{t-1}[e^{X_t}]) \right) \right] \\ &= \exp \left(\sum_{t=1}^{\tau-1} X_t - \log(\mathbb{E}_{t-1}[e^{X_t}]) \right) \cdot \mathbb{E}_{\tau-1} [\exp(X_\tau - \log(\mathbb{E}_{\tau-1}[e^{X_\tau}]))] \\ &= \exp \left(\sum_{t=1}^{\tau-1} X_t - \log(\mathbb{E}_{t-1}[e^{X_t}]) \right) \\ &= Z_\tau.\end{aligned}$$

Since $Z_0 = 1$, Ville's inequality (e.g., Howard et al. [45]) implies that for all $\lambda > 0$,

$$\mathbb{P}_0(\exists \tau : Z_\tau > \lambda) \leq \frac{1}{\lambda}.$$

The result now follows by the Chernoff method. $\square$

The next result is a martingale counterpart to Bernstein's inequality (Lemma 5).

Lemma 35 (Freedman's inequality (Bernstein for martingales)): Let $(X_t)_{t \leq T}$ be a real-valued martingale difference sequence adapted to a filtration $(\mathcal{F}_t)_{t \leq T}$. If $|X_t| \leq R$ almost surely, then for any $\eta \in (0, 1/R)$, with probability at least $1 - \delta$, for all $T' \leq T$,

$$\sum_{t=1}^{T'} X_t \leq \eta \sum_{t=1}^{T'} \mathbb{E}_{t-1}[X_t^2] + \frac{\log(\delta^{-1})}{\eta}.$$

Proof of Lemma 35. Without loss of generality, let $R = 1$, and fix $\eta \in (0, 1)$. The result follows by invoking Lemma 34 with ηX_t in place of X_t , and by the facts that $e^a \leq 1 + a + (e - 2)a^2$ for $a \leq 1$ and $1 + b \leq e^b$ for all $b \in \mathbb{R}$. $\square$

The following result is an immediate consequence of Lemma 35.

Lemma 36: Let $(X_t)_{t \leq T}$ be a sequence of random variables adapted to a filtration $(\mathcal{F}_t)_{t \leq T}$. If $0 \leq X_t \leq R$ almost surely, then with probability at least $1 - \delta$,

$$\sum_{t=1}^T X_t \leq \frac{3}{2} \sum_{t=1}^T \mathbb{E}_{t-1}[X_t] + 4R \log(2\delta^{-1}),$$

and

$$\sum_{t=1}^T \mathbb{E}_{t-1}[X_t] \leq 2 \sum_{t=1}^T X_t + 8R \log(2\delta^{-1}).$$